

DL Graded Quiz M1

Question 1

Given the following code snippet for a hypermodel:

```
def build_model(hp):  
  
    units = hp.Int("units", min_value=32,  
                  max_value=512, step=32)  
  
    activation = hp.Choice("activation",  
                           ["relu", "tanh"])  
  
    dropout = hp.Boolean("dropout")  
  
    lr = hp.Float("lr", min_value=1e-4,  
                 max_value=1e-2, sampling="log")  
  
    model = call_existing_code(units=units,  
                               activation=activation, dropout=dropout,  
                               lr=lr)  
  
    return model
```

How many hyperparameters are being tuned here?

- ☐ 3
- ☐ 1
- ☐ 4
- ☒ 2

Question 2

As we decrease model complexity, we usually do what?

- ☐ Increase the generalization gap
- ☐ Decrease the loss in training
- ☐ Decrease the generalization gap
- ☒ Incur overfitting

Question 3

Suppose you have an MLP neural network with just 1 hidden layer with dropout. The model considers 5 inputs and a single-output layer. There are 20 units in the hidden layer, and the dropout rate is 20%. How many parameters (weights + bias term) will this model train?

- ☐ 141
- ☐ 117
- ☐ 113
- ☒ 155

Question 4

The basic expression for linear learning rate decay is given by:

$$LR = \frac{LR_0}{1 + \kappa t}$$

Not counting the initial learning rate, how many hyperparameters are derived from this expression?

- ☐ Just 1, the training time κ
- ☐ Just 1, the training time t
- ☐ Just 1, the rate of decay t
- ☒ Just 1, the rate of decay κ

Question 5

If we inserted the following chunk of code into our hypermodel definition:

```
activation=hp.Choice("activation",
["relu", "sigmoid", "tanh"])

for i in range(1, hp.Int("num_layers", 1,
5)):

model.add(tf.keras.layers.Dense(units=
hp.Int("units_dense_" + str(i),
min_value=1, max_value=25, step=5),
activation=activation))
```

Which will be the activation function that the model's hidden layers use?

- ☐ ReLU
- ☐ Tanh
- ☐ The best performing one between ReLU, Sigmoid, and Tanh
- ☒ Sigmoid

Question 6

Which of the following is **true** about dropout?

- ☐ It helps to avoid overfitting.
- ☐ It causes some units in the network to never activate.
- ☐ It affects the initialization of weights and bias terms.
- ☒ It contributes to increasing the generalization gap.

Question 7

Which of the following is **true** about backtesting (in the way we have performed it so far)?

- ☐ It gives us external validity for future implementation of a strategy.
- ☐ It helps us refine the general predictive power of a deep learning model.
- ☐ It help us assess the validity of our prediction model.
- ☒ It is a very simple and cost-free check on whether our trading strategy would have worked on a specific test period.

Question 8

If we wanted to implement exponential learning rate decay in our model, our learning rate will be which of the following functional forms?

☐ $LR = LR_0 e^{\kappa t}$

☐ $LR = LR_0 e^{-\kappa t}$

☐ $LR = \frac{LR_0}{1 + \kappa t}$

☒ $LR = \frac{LR_0}{1 + e^{-\kappa t}}$

Question 9

We want to tell our hypermodel to choose between ReLU, Sigmoid, and Tanh activation functions for our model to see which one performs better. Which of the following chunks of code should we insert into the hypermodel definition?

☐ `activation=hp.Choice("activation", ["relu", "sigmoid", "tanh"])`

☐ `activation=hp.Int("activation", ["relu", "sigmoid", "tanh"])`

☐ `activation=hp.Int("act_fun", ["relu", "sigmoid", "tanh"])`

☒ `activation=hp.Choice("act_fun", ["relu", "sigmoid", "tanh"])`

Question 10

Which of the following can be used to avoid exploding/vanishing gradient problems?

☐ Xavier initialization

☐ Initializing weights to zero

☐ Initializing weights to one

☒ Stochastic gradient descent

Question 11

Suppose you have an MLP neural network with just 1 hidden layer with dropout. The model considers 15

inputs and a single-output layer. There are 10 units in the hidden layer, and the dropout rate is 30%. How many parameters (weights + bias term) will this model train?

- ☐ 120
- ☐ 196
- ☐ 176
- ☒ 171



Time left:
1:29:36

QUESTIONS

- | | | | |
|----|----|----|----|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Question 12

A small improvement over the case of a constant learning rate is to let its value become smaller as training progresses. What is this technique called?

- ☐ Learning rate decay
- ☐ Vanishing learning rate
- ☐ Sequential learning rate
- ☒ Vanishing gradient descent

Question 13

Bayesian, Random search, or Hyperband are examples of which of the following?

- ☐ Loss functions
- ☐ Hyperparameters
- ☐ Optimizers (tuners) for hyperparameter optimization
- ☒ Early-stopping algorithms

Question 14

Can we use dropout during testing?

- ☐ Yes, dropout is used during testing to keep training the model.

- ☐ Yes, dropout is always used in both training and testing phases.
- ☐ Yes, dropout makes testing more accurate by reducing overfitting.
- ☒ No, dropout is only used during training to prevent overfitting.

Question 15

In order to force the training algorithm to pay more or less attention to a certain class in our data, we can use which of the following?

- ☐ Xavier initialization
- ☐ Regularization
- ☐ Dropout
- ☒ Class weights

Question 16

Which of the following is a major difference between median stopping and curve fitting algorithms?

- ☐ Median stopping consumes more computational resources.
- ☐ Curve fitting is a method with hyperparameters.
- ☐ Curve fitting does not incorporate early-stopping.
- ☒ Median stopping usually performs better in terms of accuracy.

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